Network Security

Sr. No	LAB EXPERIMENTS	Tools
1	**Simulating and Analyzing TCP/IP Flags** **Objectives:** Simulate packet transmission with specific TCP/IP flags and observe their behavior in network communication.	Wireshark
2	**Cryptographic Key Management and Distribution** **Objective:** Learn how cryptographic keys are generated, distributed, and managed in a secure system. Study how public key infrastructure (PKI) works, and implement secure key exchange protocols.	OpenSSL, GPG, Wireshark
3	**Network Scanning Using Nmap** **Objective:** Learn how to use Nmap to perform Port scanning, OS detection, and network topology mapping (Network Discovery).	Nmap, metasploitable2 server
4	**Configuring and Testing IPsec VPN with Tunnel Mode** **Objective:** Configure an IPsec VPN using Tunnel Mode between two networks and observe secure data transmission with packet analysis.	StrongSwan or OpenSwan, VirtualBox
5	**Configuring and Testing HTTPS on a Web Server** **Objective:** Implement HTTPS on a web server and test its configuration.	Apache, Nginx server, OpenSSL, web browser
6	**Securing Email Communication with PGP/GPG** **Objective:** Learn how to secure email communications by encrypting and signing emails using PGP.	gpg
7	**DDoS Attack Simulation and Mitigation Strategies** **Objectives:** Simulate a DDoS attack in a controlled environment and implement mitigation techniques using network tools.	LOIC, HOIC, Hping3, Slowloris
8	**DNS Spoofing and DNS Security Measures** **Objectives:** Simulate a DNS spoofing attack and evaluate security practices to prevent it.	Ettercap
9	**Man-in-the-Middle (MITM) Attack Simulation** **Objectives:** Conduct an MITM attack simulation to capture unencrypted data and discuss protection mechanisms.	Ettercap
10	**Intrusion Detection Using Snort** **Objective:** Install and configure Snort to detect and log intrusion attempts on a network.	Snort, Ubuntu
11	**Firewall Configuration Using iptables** **Objectives:** Set up basic and advanced firewall rules using iptables to filter and control network traffic.	Iptables, ubuntu
12	**Configuring pfsence as a Firewall and VPN Gateway** **Objectives:** Install and configure pfsence to secure network communication and set up a VPN for encrypted access.	VirtualBox, pfsence
13	**Using UFW (Uncomplicated Firewall) for Network Security** **Objectives:** Configure UFW rules to restrict or allow specific traffic on a Linux-based system.	UWF, Ubuntu, kali
14	**Configuring SSH Tunnelling for Secure Data Transmission** **Objective**: Use SSH tunnelling to secure communication between a client and server.	OpenSSH
15	**Configuring and Attack WEP, WPA, WPA2, and WPA3 Networks** **Objectives:** Set up various wireless security protocols, analyze their encryption, and test for vulnerabilities.	Aircrack-ng Suite Airgeddon wifite